

Practice Test 12 — Answer Key

Comprehensive Final Exam Preparation (Modules 01–17)

Part A: Multiple Choice

1. **A** — All living things have cellular organization — this is universal to life
2. **B** — Cell → Tissue → Organ → Organ System → Organism is the correct hierarchy
3. **C** — The independent variable is what the scientist deliberately changes
4. **D** — A theory is well-supported by extensive, repeated experimental evidence
5. **A** — Homeostasis = maintaining internal stability despite external changes
6. **B** — Atomic number = number of protons in the nucleus
7. **C** — Covalent bonds involve sharing electrons between atoms
8. **D** — Oxygen pulls shared electrons more strongly → unequal sharing → polarity
9. **A** — pH 3 is well below 7, making it strongly acidic
10. **B** — Carbon, Hydrogen, Oxygen, Nitrogen make up ~96% of living matter
11. **C** — Dehydration synthesis removes water to join monomers into polymers
12. **D** — Carbohydrates (glucose) are the body's preferred quick energy source
13. **A** — Amino acids are the monomers of proteins (20 different types)
14. **B** — The amino acid sequence dictates how the protein folds and what it does
15. **C** — DNA and RNA are nucleic acids built from nucleotide monomers

16. **D** — Cell theory: all living things are made of cells; all cells come from pre-existing cells
17. **A** — Prokaryotes lack a membrane-bound nucleus (pro = before, karyon = nucleus)
18. **B** — Mitochondria are the "powerhouse of the cell" — site of aerobic respiration
19. **C** — Ribosomes translate mRNA into proteins (protein synthesis)
20. **D** — Plant cells have a rigid cell wall made of cellulose; animal cells do not
21. **A** — Integral membrane proteins move laterally in the lipid bilayer—the "fluid mosaic."
22. **B** — Hydrophobic tails cluster inward, away from aqueous environments.
23. **C** — Cholesterol buffers membrane fluidity across temperature changes.
24. **D** — Integral (transmembrane) proteins span from one surface to the other.
25. **A** — Hypotonic surroundings cause net water influx—red blood cells swell and may lyse.
26. **B** — Metabolism includes all chemical reactions (anabolism + catabolism).
27. **C** — Exergonic reactions release energy when bonds are rearranged.
28. **D** — Enzymes speed reactions by lowering activation energy.
29. **A** — The substrate binds at the enzyme active site.
30. **C** — ATP releases usable energy when the bond to its terminal phosphate group is broken; cells continually recycle ADP back to ATP.
31. **C** — DNA (deoxyribonucleic acid) carries the genetic instructions.
32. **D** — The four DNA bases are Adenine, Thymine, Guanine, and Cytosine.
33. **A** — In DNA, Adenine pairs with Thymine (A–T).
34. **B** — The backbone is made of alternating sugar (deoxyribose) and phosphate groups.

35. **C** — Each new DNA molecule keeps one original strand and adds one newly built strand.
36. **D** — DNA is copied during S (Synthesis) phase of interphase.
37. **A** — Prophase → Metaphase → Anaphase → Telophase (remember: PMAT).
38. **B** — Chromosomes line up at the equator of the cell during Metaphase.
39. **C** — Mitosis produces two genetically identical diploid cells.
40. **D** — Mitosis is for growth, repair, and replacement of body (somatic) cells.
41. **A** — Homeostasis is the maintenance of a stable internal environment.
42. **B** — A thermostat is a classic example of negative feedback: the response (heating) opposes the change (cold) to restore the set point.
43. **C** — Osmoregulation is specifically the regulation of water and salt balance.
44. **D** — The urinary (excretory) system, primarily the kidneys, regulates water and salt levels.
45. **A** — Chewing is mechanical digestion — physically breaking food into smaller pieces without changing its chemistry.
46. **B** — An allele is one of two or more alternate forms of a gene (e.g., T for tall, t for short).
47. **C** — Genotype is the genetic makeup (e.g., Tt); phenotype is the observable trait (e.g., tall).
48. **D** — Two identical alleles (BB or bb) = homozygous.
49. **A** — Two different alleles (Bb) = heterozygous.
50. **B** — Gregor Mendel studied inheritance using pea plants (*Pisum sativum*).

51. **C** — The skeletal system supports, protects, enables movement, stores minerals, and produces blood cells — but it does not secrete digestive hormones.
52. **D** — Red bone marrow is the site of hematopoiesis (production of red blood cells, white blood cells, and platelets).
53. **A** — Osteoclasts break down bone matrix, releasing stored calcium into the bloodstream.
54. **B** — The diaphysis is the shaft (long central portion) of a long bone.
55. **C** — The epiphyses are the rounded ends of a long bone, where joints form.
56. **D** — Skeletal muscles are attached to bones and produce voluntary movement; contraction also generates heat.
57. **A** — When muscles contract, ATP is used and heat is released as a byproduct, helping maintain body temperature.
58. **B** — Whole muscle → fascicle (bundle of fibers) → muscle fiber (single cell) → myofibril (contractile strand within the fiber).
59. **C** — The sarcomere is the repeating unit between two Z-lines; it is the basic contractile unit of striated muscle.
60. **D** — Thick filaments are composed of the motor protein myosin.
61. **A** — The CNS is the brain and spinal cord — the integration and command center.
62. **B** — The PNS consists of all nerves and ganglia outside the brain and spinal cord, connecting the CNS to the rest of the body.
63. **C** — The somatic nervous system controls voluntary (conscious) movements of skeletal muscles.
64. **D** — The sympathetic division activates the "fight-or-flight" response, increasing heart rate, dilating pupils, and redirecting blood to muscles.

65. **A** — Dendrites are the branched extensions that receive signals from other neurons or sensory receptors.
66. **B** — Prokaryotic cells lack a membrane-bound nucleus; their DNA is in the nucleoid region.
67. **C** — Most bacteria have a single circular chromosome located in the nucleoid (not enclosed by a membrane).
68. **D** — Cocci are spheres, bacilli are rods, and spirilla are spirals — the three basic bacterial morphologies.
69. **A** — Gram-positive bacteria have a thick peptidoglycan wall exposed to the outside and no outer membrane.
70. **B** — Gram-positive bacteria retain the crystal violet dye during the Gram stain procedure and appear purple/violet.
71. **C** — Deoxygenated blood from the systemic veins drains into the **right atrium** first (superior and inferior vena cava), before passing to the right ventricle.
72. **D** — The right ventricle ejects blood into the **pulmonary trunk**, which branches into the **pulmonary arteries** carrying blood to the lungs.
73. **A** — The first heart sound (“lub,” S1) is mainly the closing of the **AV (atrioventricular) valves** between atria and ventricles at the start of ventricular systole.
74. **B** — The second heart sound (“dup,” S2) is mainly the closing of the **semilunar valves** (pulmonary and aortic) at the end of ventricular systole.
75. **C** — **Arteries** carry blood **away** from the heart (whether oxygen-rich or oxygen-poor depends on which artery).
76. **D** — Producers capture light energy and build organic molecules that feed other trophic levels.
77. **A** — Energy and biomass typically pass producer → primary consumer → higher consumers.

78. **B** — Biotic factors are living; abiotic factors are non-living chemical/physical conditions.
79. **C** — Materials recycle through biogeochemical cycles; energy dissipates as heat.
80. **D** — Carrying capacity (K) reflects limiting resources and space.
81. **A** — Selection requires variation, inheritance, and differences in survival/reproduction.
82. **B** — Homologous traits descend from a common ancestor; analogous traits converge by similar roles.
83. **C** — Chance allele-frequency shifts matter most when sampling error is large (small N).
84. **D** — Barriers to gene flow allow divergence—prezygotic or postzygotic mechanisms.
85. **A** — Transitional fossils with intermediate body plans (e.g., *Tiktaalik*) align with descent with modification from shared ancestors.
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Part B: Fill in the Blank

86. Accept **observation** (also Observation, experimental observation).
87. Accept **organism** (also organism).
88. Accept **covalent** (also covalent).
89. Accept **acidic** (also acidic, acid).
90. Accept **protein** (also protein).
91. Accept **nitrogenous** (also nitrogenous).
92. Accept **mitochondria** (also mitochondria, mitochondrion).
93. Accept **nucleus** (also nucleus).
94. Accept **down** (also down).

95. Accept **osmosis** (also osmosis).
96. Accept **respiration** (also respiration , cellular respiration).
97. Accept **glucose** (also glucose).
98. Accept **template** (also template).
99. Accept **mRNA** (also mRNA , messenger RNA).
100. Accept **anaphase** (also anaphase).
101. Accept **meiosis** (also meiosis).
102. Accept **negative** (also Negative).
103. Accept **small** (also small).
104. Accept **heterozygous** (also heterozygous).
105. Accept **sex** (also sex , X and Y , X/Y).
106. Accept **osteoclasts** (also osteoclasts , osteoclast).
107. Accept **synovial** (also synovial).
108. Accept **myosin** (also myosin).
109. Accept **fibers** (also fibers , fibres).
110. Accept **potentials** (also potentials , potential).
111. Accept **cleft** (also cleft).
112. Accept **membrane** (also membrane).
113. Accept **plasmids** (also plasmids , plasmid).
114. Accept **aorta** (also aorta).

115. Accept **capillaries** (also capillaries, capillary).

116. Accept **trophic** (also trophic).

117. Accept **resources** (also resources, resource).

118. Accept **variation** (also variation).

119. Accept **isolation** (also isolation, reproductive isolation).

Part C: Short Answer / Free Response

120. Sample points for full credit: - Observation records measurable evidence; opinion states belief without required evidence.

121. Sample points for full credit: - Stable temperature/pH/blood chemistry keeps enzymes near optimal activity ranges.

122. Sample points for full credit: - Ionic: electron transfer forming ions attracted together (NaCl). Covalent: shared pairs (water O–H).

123. Sample points for full credit: - Lower pH means higher hydrogen ion concentration—more acidic.

124. Sample points for full credit: - Specific 3-D active site fits substrates; denatured enzyme loses catalytic shape.

125. Sample points for full credit: - Linear/base-paired sequences encode digital-like instructions sugars alone lack.

126. Sample points for full credit: - Mitochondria oxidize fuel aerobically; chloroplasts trap light energy in plants/algae.

127. Sample points for full credit: - Both translate mRNA with ribosomal machinery—universal genetic code.

- 128.** Sample points for full credit: - Passive: diffusion/facilitated down gradients; Active: pumps move against gradients (Na^+ / K^+ pump).
- 129.** Sample points for full credit: - Hypertonic medium draws water out—cells shrink (crenation for RBCs).
- 130.** Sample points for full credit: - Maintenance transport, biosynthesis, signaling, repair—even Basal metabolic demand consumes ATP.
- 131.** Sample points for full credit: - Aerobic uses oxygen as final electron acceptor; fermentation regenerates NAD^+ without O_2 .
- 132.** Sample points for full credit: - DNA replication/transcription → RNA processing/export → translation → folded protein.
- 133.** Sample points for full credit: - Mutation is DNA sequence change—provides variation raw material for evolution.
- 134.** Sample points for full credit: - Mitosis clones somatic genome; Meiosis halves homologues for gametes.
- 135.** Sample points for full credit: - Chiasmata exchange segments—new allele combos on chromosomes.
- 136.** Sample points for full credit: - Examples: epithelium lining airways; bone/cartilage connective support; skeletal muscle moves bones.
- 137.** Sample points for full credit: - Cartilage flexible matrix resists compression at joints; rigid bone handles torque.
- 138.** Sample points for full credit: - Dominant healthy allele masks recessive disease allele in heterozygotes.
- 139.** Sample points for full credit: - Pedigrees show inheritance patterns across families; Punnett predicts offspring ratios.
- 140.** Sample points for full credit: - Osteoclasts remove damaged bone; osteoblasts lay new matrix—Wolff's law adaptation.

- 141.** Sample points for full credit: - Synovial fluid secreted into cavity lubricates cartilage surfaces.
- 142.** Sample points for full credit: - Ca^{2+} binds troponin → tropomyosin shift exposes actin sites for myosin.
- 143.** Sample points for full credit: - More motor units (and larger ones) are activated as needed; recruiting additional units increases the number of muscle fibers contracting and therefore the total force produced.
- 144.** Sample points for full credit: - Resting negative inside becomes less negative during Na^{+} influx—depolarization phase.
- 145.** Sample points for full credit: - Sympathetic raises HR/stress; parasympathetic lowers HR/digest emphasis.
- 146.** Sample points for full credit: - Full dosing kills susceptible bacteria—less chance survivors mutate resistance.
- 147.** Sample points for full credit: - Bacteria metabolize independently divide; viruses need host machinery.
- 148.** Sample points for full credit: - Pulmonary: right heart→lungs for oxygen pick-up; Systemic: left heart→body delivery.
- 149.** Sample points for full credit: - In the alveoli, oxygen (O_2) diffuses from inhaled air into the blood, and carbon dioxide (CO_2) diffuses from the blood into the alveolar air to be exhaled. Each gas moves down its concentration gradient across the thin alveolar–capillary membrane.
- 150.** Sample points for full credit: - Energy enters as sunlight→ producers → herbivores → carnivores; nutrients cycle biogeochemically.
- 151.** Sample points for full credit: - K reflects limits—factors like drought/predation reduce carrying capacity.

152. Sample points for full credit: - Traits inherited vary; some reproduce more—allele frequencies shift generations.

153. Sample points for full credit: - Examples: temporal mating seasons, geographic barriers, gametic incompatibility reducing gene flow.